



AURORA ROBOTICS

Computer Vision for Autonomous Vehicles

A Mathematical Companion

Lecture 1 — Images, Pixels, and the Building Blocks

How to use this companion

- ▶ Every section follows the lecture slides — in the same order.
- ▶ Each equation is followed by a plain-English reading.
- ▶ Each topic ends with a worked example you can do by hand.
- ▶ Numerical tolerances: results match to $\leq 10^{-3}$.



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